Multi-armed bandits with arriving arms: the cost of reliable set maintenance

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Abstract

Sequential experiments may acquire new alternatives while earlier alternatives remain available. A procedure can accumulate little regret and still provide little evidence about which alternatives are currently near optimal. We study the joint cost of reward collection and reliable maintenance of the complete relative near-optimal set. Reports are correct simultaneously over arms and time; timeliness is measured by cumulative unresolved arm time outside a boundary band. We show that uncertainty about one arrival can obstruct positive certification of many earlier arms. This yields an arrival-growth boundary for sublinear maintenance area, and the worst-schedule area is characterized up to universal constants. A repeated-update construction separates prompt classification from permitted deferral: at a specified horizon, the former requires regret proportional to the inverse squared boundary distance, whereas the latter admits only logarithmic dependence when the other parameters are fixed. The necessary cost also holds with fixed positive probability under the same maintenance requirement. A checkpoint policy reuses observations across arrivals and supplies executable upper bounds. Comparisons with retained-history confidence methods show both the reporting cost missed by regret alone and the practical conservatism of the checkpoint construction.

Keywords: adaptive experimentation; all-good-arm identification; arriving arms; cumulative unresolved time; sequential inference.

1 Introduction

A sequential experiment may need to retain several promising alternatives while new alternatives continue to enter. Examples include a testing platform that receives externally generated designs and an experimental pipeline that introduces candidates during an ongoing study. The investigator must decide which alternative to sample and what the accumulated evidence says about the whole

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candidate set. Cumulative reward loss measures the quality of allocation, but a small loss does not guarantee that all alternatives within a specified tolerance of the current best can be distinguished from those outside it.

Consider a collection of well-studied candidates and one newly available arm. Although the earlier means have not changed, uncertainty about the new arm's possible superiority may prevent positive certification of many earlier candidates. An informative upper bound on its mean may be needed before the earlier candidates can be certified. A candidate close to today's relative threshold may itself require many observations to classify. After a sufficiently better arm arrives, both possible sides of the old threshold can lie well below the new one. Observations already collected can then suffice to exclude the candidate, without resolving its earlier membership. Continual reporting therefore creates shared waiting costs, while later arrivals can make unfinished fine-resolution tasks unnecessary.

We study these effects in the stationary stochastic bandit model with exogenous permanent arrivals. Regret is measured against the best arm available at each physical observation time. This model and comparator are established in ballooning bandits (**?**). For any finite prefix, permanent arrival is also a monotone availability pattern in the sleeping-bandit formulation (**?**). An ordinary index policy can admit an arm on arrival. The additional statistical requirement considered here is a reliable report of the complete current relative near-optimal set, together with a bound on how long its members and nonmembers remain unresolved.

The fixed-arm version of relative-good-set identification is the ALL-ε problem of **?**. Their analysis already distinguishes discrimination near the membership boundary from uncertainty about the best mean. Their algorithms, including $(\mathrm{ST})^2$ and FAREAST, are essential comparisons for any proposed extension. Reusing confidence bounds after an arrival is a natural adaptation; the cost of maintaining their reports through repeated changes of the relative threshold requires further analysis. Our lower bounds measure cumulative unresolved arm time under one simultaneous correctness requirement. They concern continuing reports, rather than a single terminal recommendation or independently imposed deadlines. Fixed-budget thresholding also studies terminal classification loss (**?**); cumulative unresolved time records the full reporting path.

Joint reward and identification objectives also have substantial precedents. **?** study the compromise between regret and decision time; **?** characterize regret-optimal fixed-confidence best-arm identification and its sampling cost. **?** combine reward-oriented sampling with anytime-valid testing for fixed-threshold good-arm identification. Separating a reward policy from a reporting layer thus has established precedents. Sequential multiple testing and support recovery likewise supply the simultaneous-recovery scales used below (**??**). We study how a continuing obligation to report the relative set imposes these costs, shares them among arms, and sometimes allows them to be avoided.

Our first main result gives a common-environment lower bound: an insufficiently studied arrival can keep a whole cohort unresolved. Combined with an executable upper bound, it identifies the polynomial arrival exponent below which both expected regret and cumulative unresolved area can be sublinear. A separate worst-schedule result characterizes the area as the smaller of the available

arm-time and a quadratic-in-arm-count identification cost. Its lower bound already occurs for an initial batch. Repeated benchmark improvements have a further temporal effect, which we analyze separately.

Our second main result isolates the cost of temporary membership questions. In an explicit multiple-update family, a small area budget forces enough fine-resolution classifications before they expire. At a specified sequence of horizons this gives matching rapid-maintenance regret bounds. A larger constant-factor area budget permits coarse evidence to be reused after the next improvement and has a substantially smaller attainable regret. We also prove a fixed-probability regret lower bound under precisely the same area requirement. These results describe two service budgets; they do not identify the entire instancewise frontier.

The checkpoint construction is deliberately simple. It accepts arms immediately, retains all observations, and interleaves reporting service with full-history UCB. It provides a constructive bound whose reward guarantee holds without conditioning on correct reports. Its limitations are visible in the numerical comparisons: strong confidence-based adaptations can improve both regret and area, and fine fixed checkpoints can be impractical at moderate budgets. The construction establishes attainable joint costs and explains their mechanism; it need not be the best practical implementation.

2 Observation model and reporting requirement

Arms are indexed by arrival order, with a fixed order within a batch. Arm i arrives at a deterministic time τ_i and remains available. Let

$$A_t = \{i : \tau_i \le t\}, \qquad K_t = |A_t| < \infty.$$

We assume $A_1 \neq \varnothing$. Arm i has an independent potential reward stream $(X_{i,n})_{n\ge1}$ of independent, identically distributed observations in $[0,1]$, with mean μ_i. At round t the policy sees current arrivals, chooses $I_t \in A_t$ from the observed history and independent randomization, and observes the next entry of that arm's stream. It then issues its report. The policy does not know means, future arrivals, or the evaluation endpoint. Conditioning on a reward-independent exogenous random schedule gives the same conditional guarantees when that extension is needed.

Write $N_i(t)$ for lifetime sample counts and $\widehat{\mu}_{i,n}$ for the mean of the first n potential observations. Define

$$B_t = \max_{i\in A_t} \mu_i, \qquad \Delta_i(t) = B_t - \mu_i, \qquad R_T = \sum_{t=1}^{T} \Delta_{I_t}(t).$$

Here R_T is random pseudo-regret; $\mathbb{E}R_T$ is its expectation. All endpoints T below refer to actual physical pulls.

For $0 < \varepsilon < 1$, the target is

$$G_t^{\varepsilon} = \{i \in A_t : \Delta_i(t) \le \varepsilon\}.$$

A report partitions A_t into C_t, D_t, V_t: certified good, certified bad, and unresolved. With $0 < \delta < 1/2$,

let

$$E_{\mathrm{lab}} = \{\text{for all } t:\ C_t \subseteq G_t^{\varepsilon} \subseteq A_t \setminus D_t\}.$$

A policy is δ-reliable if $\mathbb{P}(E_{\mathrm{lab}}) \geq 1-\delta$ in every allowed reward environment. Positive reports may need revision after an arrival. A sound negative report remains valid because B_t is nondecreasing. This statement refers to the model's stationary means and permanent availability.

Reliability alone permits every arm to remain unresolved. To measure progress, fix a boundary width $0 < \eta < 1$ and define

$$d_i^{\varepsilon}(t) = |\Delta_i(t) - \varepsilon|, \quad U_t(\eta) = \sum_{i \in A_t} \mathbf{1}\{i \in V_t, d_i^{\varepsilon}(t) > \eta\}, \quad \mathcal{A}_T(\eta) = \sum_{t=1}^{T} U_t(\eta). \tag{1}$$

An arm within the boundary band still requires a correct label whenever a label is issued; its unresolved time is simply exempt from the service loss. No assumption removes such arms from the model. The parameters have different roles: ε defines membership, η defines the precision at which unresolved time is charged, and a policy's service precision will be denoted by ρ.

The main joint requirement at a fixed T is

$$\mathbb{E}_{\theta} R_T \leq r, \qquad \mathbb{P}_{\theta}\{E_{\mathrm{lab}}, \mathcal{A}_T(\eta) \leq a\} \geq 1-\delta. \tag{2}$$

It uses an expected reward cost and a joint high-probability reporting cost. It is stronger than control of per-arm errors, and differs from false discovery rate control. In particular, $\mathcal{A}_T = o(T)$ requires the time-average *number* of unresolved separated arms to vanish. It does not merely require a small unresolved fraction among all arrived arms.

For a performance family Θ, write $\mathfrak{R}_{\Theta}(a)$ for the infimum of $\sup_{\theta \in \Theta} \mathbb{E}_{\theta} R_T$ over policies reliable on the full bounded model and satisfying the second condition of (**??**) for every $\theta \in \Theta$. The value is $+\infty$ if no policy is feasible. The lower bounds remain valid if reliability is required only on the lower-bound subfamily.

2.1 Admission and retirement in the reward account

The distinction between statistical reports and reward eligibility is useful even before introducing a policy. Let $S_t \subseteq A_t$ be an admitted set and $F_t \subseteq S_t$ its nonempty reward-active subset, with $I_t \in F_t$. At every deterministic endpoint, including one inside an algorithmic round,

$$\begin{aligned} R_T = &\sum_{t \leq T} (\max_{i \in F_t} \mu_i - \mu_{I_t}) + \sum_{t \leq T} (B_t - \max_{i \in S_t} \mu_i) \\ &+ \sum_{t \leq T} (\max_{i \in S_t} \mu_i - \max_{i \in F_t} \mu_i). \end{aligned} \tag{3}$$

This follows by adding and subtracting the two intermediate maxima at each pull. The terms are within-active learning, admission delay, and loss of an admitted benchmark through retirement. The latter two cannot in general be identified with arrival increments alone. Our checkpoint policy uses $F_t = S_t = A_t$ for reward selection. Its reporting exclusions consequently do not create persistent

loss of a reward benchmark.

3 Lifetime checkpoints and reward selection

Choose $0 < \rho < 1$ before observing rewards. For arm i, set

$$q_i(\rho) = \left\lceil \frac{2}{\rho^2} \log \frac{4i(i+1)}{\delta} \right\rceil. \tag{4}$$

Before this count is reached, its reporting interval is $[0, 1]$. Afterwards, the interval is permanently

$$[\ell_i, u_i] = [\widehat{\mu}_{i,q_i} - \rho/2, \widehat{\mu}_{i,q_i} + \rho/2] \cap [0, 1]. \tag{5}$$

The interval uses a fixed potential prefix, even though its completion time is adaptive. Later observations continue to update the reward estimate. One can also prescribe different precisions ρ_i before observing the corresponding arms; selecting a favorable checkpoint from their realized data is not covered.

CP-UCB-AA uses the following physical-clock rule.

1. Admit current arrivals and append their checkpoints to a queue in arrival order. Initialize their counts to zero.

2. At an even round, if a checkpoint is incomplete, sample the first incomplete arm in the queue. These are designated service pulls.

3. At every other round maximize, over all $i \in A_t$,

$$J_i(t) = \widehat{\mu}_{i,N_i(t-1)} + \sqrt{\frac{3\log(2t)}{N_i(t-1)}}. \tag{6}$$

   Use $+\infty$ for a zero count and break ties in arrival order.

4. Update the chosen arm's lifetime count and reward sum. If the count first equals q_i, construct (**??**). Every observation contributes to checkpoint completion, including observations from (**??**).

5. Set $\underline{B}_t = \max_{i \in A_t} \ell_i$ and $\overline{B}_t = \max_{i \in A_t} u_i$, and report

$$C_t = \{i : \ell_i \geq \overline{B}_t - \varepsilon\}, \qquad D_t = \{i : u_i < \underline{B}_t - \varepsilon\}, \qquad V_t = A_t \setminus (C_t \cup D_t). \tag{7}$$

The intervals are nonempty on every sample path, so C_t and D_t cannot overlap. All arrived arms remain eligible for (**??**), including those in D_t. The method stores a count, sum, and reporting interval for every arm; it does not claim to reduce total arm storage. A direct index or report update costs $O(K_t)$ operations. The executable fixed-precision reference is supplied with the numerical code.

Proposition 3.1 (Simultaneous reporting and unconditional reward cost)**.** *On the full bounded model, CP-UCB-AA satisfies* $\mathbb{P}(E_{\mathrm{lab}}) \geq 1 - \delta/2$*. If* $\mathcal{I}_T$ *is its set of designated service rounds, then at every deterministic* T*,*

$$\mathbb{E}R_T \leq \min\left\{T, K_T + 4\sqrt{3K_T T \log(2T)} + 1 + \mathbb{E}\sum_{t\in\mathcal{I}_T} \Delta_{I_t}(t)\right\}. \tag{8}$$

Pathwise, $|\mathcal{I}_T| \leq \sum_{i\leq K_T} q_i$*.*

Hoeffding's inequality bounds the failure at checkpoint i by $\delta/\{2i(i+1)\}$. Summing over potential arm indices gives one coverage event of probability at least $1-\delta/2$, on which every report is correct. For (**??**), a separate, summable concentration argument at physical time t controls ordinary UCB pulls. Each arm and pre-pull count is charged at most once; inserted service observations only skip counts. This proof holds on all reporting paths and therefore incurs no omitted δT term. Complete proofs appear in the supplementary material.

The construction retains the screening motivation of UCB-AA but changes its algorithmic clock and its statistical task. It has no prescribed terminal round, no delayed admission, and no restart that is completed artificially for analysis. The guarantees above concern this construction, not the original round-based elimination algorithm.

4 The burden of continual certification

4.1 One uncertain arrival can obstruct many reports

The first lower bound retains one reward environment throughout the time axis. It therefore measures a cumulative burden without adding the worst cases of incompatible experiments. Fix $0 < \varepsilon \leq 1/16$, $0 < \delta \leq 1/16$, and $0 < \eta < \varepsilon/2$. An initial reference has Bernoulli mean $1/4+\varepsilon/2$. Candidates arrive on any prescribed schedule. In environment zero each has mean $p = 1/4$; environment j changes only candidate j to mean $p + 2\varepsilon$. All candidates are good in environment zero, with boundary distance $\varepsilon/2$. After candidate j arrives in environment j, every other candidate is bad.

Let M_t count arrived candidates and $a_t = M_t - M_{t-1}$. Define

$$q_- = \left\lfloor \frac{\mathrm{kl}(1/4, \delta)}{\mathrm{kl}(1/4, 1/4 + 2\varepsilon)} \right\rfloor, \qquad Q_t = (Q_{t-1} + q_- a_t - 1)_+, \quad Q_0 = 0,$$
$$\mathcal{B}_T = \sum_{t\leq T} (M_t - 1)_+ \mathbf{1}\{Q_t > 0\}. \tag{9}$$

The queue is an information-capacity certificate: each arriving candidate requires q_- units, while only one observation is possible per round. It is not the queue of any particular policy. In the stated range $q_- \geq 2$.

Theorem 4.1 (Broadcast uncertainty)**.** *For every policy reliable on environment zero and all its one-candidate alternatives,*

$$\mathbb{E}_0 \mathcal{A}_T(\eta) \geq (3/4 - \delta)\mathcal{B}_T.$$

If $\mathcal{B}_T > 0$, then

$$\mathbb{P}_0\{\mathcal{A}_T(\eta) \geq \mathcal{B}_T/4\} \geq 1/4.$$

These conclusions still hold when the policy knows the full arrival schedule, the horizon, and the reference distribution.

At a busy time, some deterministic candidate label has expected sample count below q_-. Changing that candidate alone has too little history KL to permit positive certification of another candidate with probability at least $1/4$. Reliability also bounds incorrect negative reports. The remaining probability must be unresolved. The same reference environment supports this argument at every time; the difficult label is selected from expected counts rather than from a favorable realized history. The supplementary proof includes the bounded-variable step needed for the probability statement.

If m candidates have arrived by time s, the queue obeys the deterministic inequality

$$\sum_{t \leq s+qm-1} (M_t - 1)_+ \mathbf{1}\{Q_t(q) > 0\} \geq \frac{q-1}{2} m(m-1), \qquad q \geq 2. \tag{10}$$

The factor m^2 reflects the number of affected reports as well as the work needed to exclude high-quality alternatives. The subtraction of one unit in $q-1$ accounts for the last observation of each busy period: reports are issued after service.

Corollary 4.2 (Arrival-growth boundary)**.** *Fix the preceding task parameters and one infinite arrival schedule. If a policy is reliable for every reward environment and has a deterministic envelope $a_T = o(T)$ satisfying (**??**) in every environment, then $K_t = o(\sqrt{t})$. Conversely, $K_T^2 \log(eK_T) = o(T)$ suffices for one horizon-free policy to have both $\mathbb{E}R_T = o(T)$ and such a sublinear area envelope. In particular, for $K_t \asymp t^\alpha$, $0 < \alpha < 1$, simultaneous attainment is possible if and only if $\alpha < 1/2$.*

The sufficiency follows from the next theorem. The exponent statement leaves a logarithmic gap for general critical schedules. It does not identify $K_t = o(\sqrt{t})$ as a sufficient condition on every schedule.

4.2 Worst-schedule maintenance area

Let $\mathfrak{A}_{T,K}$ be the smallest a attainable by a full-model reliable policy with

$$\sup_{\tau: K_T \leq K} \sup_{\mu} \mathbb{P}_\mu\{E_{\mathrm{lab}}^c \text{ or } \mathcal{A}_T(\eta) > a\} \leq \delta.$$

Policies may know T, K and the schedule for the lower bound. The upper bound uses neither T nor K as input.

Theorem 4.3 (Worst-schedule area and joint attainability)**.** *For $K \geq 2$, $T \geq 1$, $0 < \varepsilon \leq 1/8$, $0 < \eta \leq \varepsilon/32$, and $0 < \delta \leq 1/16$,*

$$2^{-24} \min\left\{KT, \frac{K^2}{\eta^2} \log \frac{K}{\delta}\right\} \leq \mathfrak{A}_{T,K} \leq 256 \min\left\{KT, \frac{K^2}{\eta^2} \log \frac{K}{\delta}\right\}. \tag{11}$$

The CP policy with $\rho = \eta/4$ attains the upper order and also satisfies

$$\mathbb{E}R_T \le \min\left\{T, K + 4\sqrt{3KT\log(2T)} + 1 + 99K\eta^{-2}\log(K/\delta)\right\}. \tag{12}$$

For the upper bound, completed checkpoints imply endpoint errors at most $\eta/4$, sufficient to classify all arms outside the boundary band. The unfinished-work queue is dominated by adding q_i at each arrival and serving at even rounds. On the simultaneous coverage event,

$$\mathcal{A}_T(\eta) \le \min\left\{K_T T, 2K_T \sum_{i \le K_T} q_i\right\}. \tag{13}$$

Additional UCB observations reduce the work; they are not charged again as new checkpoint samples. The lower bound uses $K-1$ candidates with independent means $1/2 \pm 2\eta$ and a reference of mean $1/2+\varepsilon$, all available at the first pull. Simultaneous partial recovery and a horizon truncation give the probability statement required by the definition of $\mathfrak{A}_{T,K}$. The statistical $\log K$ factor has fixed-arm multiple-testing precedents; the lower construction itself requires no repeated arrivals.

For fixed task parameters, (**??**) implies a precise uniform terminal-count statement: over all schedules with $K_T \le k(T)$, sublinear expected regret and sublinear service envelopes are jointly attainable if and only if $k(T)^2 \log k(T) = o(T)$. Here the worst schedule can change with T. On a single fixed schedule that conclusion cannot be inferred from total growth alone. For example, batches of size

$$m_n = \left\lfloor \frac{\sqrt{s_n}}{(\log s_n)^{1/4}} \right\rfloor, \qquad s_n = 2^{2^n},$$

give $K_t = o(\sqrt{t})$ but preclude any $o(T)$ area envelope uniform over reward environments. Near s_n, simultaneous classification of that burst requires accumulated unresolved area of order $m_n^2 \log m_n$, which exceeds s_n by a growing factor. The supplementary argument selects a single infinite schedule; the difficult reward environment is allowed to vary with the endpoint, as required by the stated uniform guarantee.

5 Temporary membership questions and their regret cost

5.1 A repeated-update family

We now keep the reward environment, arrival schedule, endpoint, and reporting parameters fixed when comparing two area budgets. Choose integers $J \ge 1$, $M \ge \max\{2, J\}$ and

$$0 < \varepsilon \le 1/8, \quad 0 < \delta \le 1/16, \quad Jv \le \varepsilon/4, \quad 0 < d \le v/32, \quad \eta = d/2. \tag{14}$$

At $a_e = 1 + eH$, $e = 0, \dots, J$, a reference with mean $b_e = 1/2 + ev$ arrives. For each $e < J$, the same batch contains M candidates with Bernoulli means

$$\mu_{e,k} = b_e - \varepsilon + \sigma_{e,k} d, \qquad \sigma_{e,k} \in \{-1, +1\}. \tag{15}$$

The last batch contains only its reference, and $T = (J+1)H$. The family Θ includes every sign configuration and every within-batch permutation. The policy is not told which arm is the reference. Lower bounds may disclose those identities and the schedule. All arms remain available.

In its birth window a candidate is good exactly when its sign is positive. Its regret gap is approximately ε, but its distance to the membership boundary is d. At the next update both signs become bad, with boundary distance at least $v - d$. Every reference remains good. Thus the final good set consists only of references, regardless of all the candidate signs. The statistical tasks that distinguish the environments are temporarily relevant and can expire while their arms remain available.

5.2 Area forces completion of enough temporary tasks

The lower-bound ingredient permits arbitrary interleaving and stopping. For m independent uniform signs with observation laws P_i^+, P_i^-, put

$$\kappa = \max_i \max\{D(P_i^+ \| P_i^-), D(P_i^- \| P_i^+)\} \in (0, \infty).$$

If a bounded stopping time has at least half the signs determined with prior probability at least $1/2$, and all configurations have familywise report error at most $\delta \le 1/16$, then

$$\overline{\mathbb{E}} \sum_{i=1}^{m} N_i \ge \frac{m}{1024\kappa} \log \frac{m}{\delta}. \tag{16}$$

Other observations must not encode these signs. The proof factors the posterior after the sampling and stopping kernels cancel, and combines a simultaneous recovery bound with coordinatewise change of measure (**?**). It does not assert independence of the random sample counts. The logarithmic simultaneous-recovery scale is established statistical structure; its role here is to price partially completed, expiring tasks.

For clarity, a heterogeneous version gives an explicit lower allocation constraint. Let a compatible batch e have M_e candidates, window length H_e, boundary distance $d_e > \eta$, and two-way KL bounded by κ_e. Put $w_e = M_e H_e$, $V = \sum_e w_e$. Define

$$\Phi_\delta(u) = \begin{cases} \mathrm{kl}(1 - \delta - u, \delta), & u < 1 - 2\delta, \\ 0, & u \ge 1 - 2\delta, \end{cases}$$

$$g_{m,\delta}(p) = \max\left\{0, \frac{(p - 2\delta)_+}{2}\left[\log \frac{m}{4 \log 2}\right]_+ - 1, \Phi_\delta(1 - p/2)\right\}.$$

If a policy satisfies the joint area requirement with budget a, then

$$\sup_\sigma \mathbb{E}_\sigma R_T \ge \inf_{\substack{0 \le p_e \le 1 \\ \sum_e w_e p_e \ge (1-\delta)V - 2a}} \sum_e \frac{(\varepsilon - d_e) M_e}{\kappa_e} g_{M_e,\delta}(p_e). \tag{17}$$

Here p_e is the probability of determining at least half of batch e before its window ends. A window that never does so contributes at least $w_e/2$ area. This gives the constraint directly from the area success event, with its failure probability retained. The objective then uses (**??**) under one compatible product prior. The policy can choose where to defer; no separate requirement forces every window to finish. The relaxation need not be attainable in general.

5.3 Reusable coarse evidence and a matched rapid endpoint

CP-UCB-AA can use a predetermined precision ρ_e for each arriving batch. Let $S_e = \sum_{i:\tau_i=a_e} q_i(\rho_e)$ be the total checkpoint work of batch e, and set $M_{-1} = M_J = 0$. In a heterogeneous repeated-update family, suppose

$$H_e \geq 2S_e, \qquad b_J - b_0 \leq \varepsilon/4, \qquad d_e + \rho_e + \rho_{e+1} < b_{e+1} - b_e, \qquad 2\max_e \rho_e \leq \varepsilon/2. \tag{18}$$

These are conditions on the performance family, not future inputs supplied to the policy. Write $\mathcal{F} = \{e < J : 2\rho_e < d_e\}$.

Proposition 5.1 (Shared update from lifetime evidence)**.** *On the checkpoint coverage event, under* (**??**)*,*

$$\mathcal{A}_T(\eta) \leq \sum_{e\notin\mathcal{F}} M_e H_e + 2\sum_{e\in\mathcal{F}} M_e S_e + 2\sum_{e=0}^{J}(e+1+M_{e-1})S_e. \tag{19}$$

When the next batch completes its checkpoints, every preceding-batch candidate can be excluded using its saved interval and the new reference lower bound, without another observation of that old candidate for certification.

The last term charges only temporarily affected references and the preceding candidate cohort, rather than all historical candidates at each update. The conclusion follows because the new lower bound exceeds every preceding candidate upper bound by more than ε. The condition $H_e \geq 2S_e$ ensures that all its checkpoint observations occur in its birth window. The corresponding unconditional reward bound, including excess UCB pulls, is given in the supplement.

For the homogeneous family (**??**)–(**??**), set

$$K = J(M+1)+1, \quad L_K = \log\frac{4K(K+1)}{\delta}, \quad \rho_f = d/4, \quad \rho_c = v/8,$$
$$q_f = \lceil 32L_K/d^2\rceil, \quad q_c = \lceil 128L_K/v^2\rceil, \quad H = 128(M+J+1)q_f, \quad V = JMH. \tag{20}$$

The quantities q_f, q_c are uniform analysis bounds on the actual armwise quotas. Neither final arm count nor future times are policy inputs. Fine precision can be set from the task input as $\rho_f = \eta/2$; the coarse precision is a separate preselected configuration. Automatic adaptation to an unknown improvement scale is not asserted.

Theorem 5.2 (Rapid and deferred maintenance at the same endpoint)**.** *On the preceding family and horizon,*

$$2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta} \leq \mathfrak{R}_\Theta(V/8) \leq 2^{18}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}. \tag{21}$$

The coarse-precision CP policy has joint label-and-area success probability at least $1-\delta$ *with area budget* $9V/8$*, and*

$$\sup_{\theta\in\Theta}\mathbb{E}_\theta R_T \le 2\varepsilon JMq_c + 64\left(\frac{JM}{\varepsilon}+\frac{J^2}{v}\right)\log(2T)+1. \tag{22}$$

Consequently, at fixed legal $J, M, \varepsilon, v, \delta$ *and* $d \downarrow 0$*, with* $\eta = d/2$ *and the horizon in* (**??**)*,* $\mathfrak{R}_\Theta(V/8) = \Theta(d^{-2})$*, whereas* $\mathfrak{R}_\Theta(9V/8) = O(\log(1/d))$*.*

For the lower bound, area at most $V/8$ forces half-completion in at least $3J/4$ windows on success. Enough windows then require (**??**), and each candidate observation costs at least $\varepsilon - d$. Distinct batches' samples are charged once. For the upper bound, (**??**) gives the appropriate area budgets. At the specified horizon the deterministic bound $R_T \le 2\varepsilon T$ suffices for the rapid order; it would not justify the same conclusion at arbitrarily longer windows. The coarse bound instead counts checkpoint observations and excess ordinary UCB pulls separately.

The two budgets differ by a constant factor, but they allow different service levels. The theorem does not claim that the coarse policy meets the rapid budget, that its regret upper bound is matched, or that all intermediate budgets are characterized. A preselected mixture of fine and coarse batches gives additional attainable points, as recorded in the supplement.

5.4 A probability lower bound under the same area requirement

An expectation lower bound alone could, in principle, reflect unusual sample paths. The following result gives a stronger conclusion without replacing the cumulative area requirement by deadlines.

Theorem 5.3 (A nonvanishing probability of the rapid-maintenance cost)**.** *In the repeated-update family, suppose a reliable policy satisfies*

$$\mathbb{P}_\sigma\{E_{\mathrm{lab}}, \mathcal{A}_T(\eta) \le V/8\} \ge 1-\delta$$

for every sign configuration. Then one fixed configuration satisfies

$$\mathbb{P}_\sigma\left\{R_T \ge 2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}\right\} \ge \frac{11}{80}. \tag{23}$$

The proof stops each window at half-completion, its endpoint, or the first candidate sample beyond a fixed budget. Applying the partial-recovery lemma to this stopped experiment bounds the probability of a cheap completion by $1/2$. Low total area requires many completions; a count of cheap completions then leaves enough expensive windows with probability at least $1/5 - \delta \ge 11/80$. Their actual observations give (**??**). No independence between window-completion events is needed. Since $11/80 > \delta$, the rapid endpoint in (**??**) also has matching order when regret and area are bounded on one event of probability at least $1-\delta$. The full stopping argument and a compatible multiple-window area truncation are proved in the supplement.

6 Numerical comparisons at a fixed pull budget

We examine the cost of maintaining reliable reports at an actual sampling budget, including settings in which the conservative checkpoint construction is impractical. These comparisons assess finite-budget behavior. The four environments are not the separated family used for the two-endpoint theorem, and the simulations do not test its asymptotic rates.

6.1 Protocol and competing rules

All rewards are Bernoulli. We fix $T = 20{,}000$, $\varepsilon = .30$, $\eta = .10$, and $\delta = .05$, with eight arms by the end of each run. There are 160 independent replications per environment. Within a replication, every method receives the same potential reward stream for each arm; its nth pull of arm i reveals the same $X_{i,n}$. Thus methods share the underlying observations without being forced to take the same actions. Arrivals occur before the action, reports follow the observation, and all methods retain previous observations. The full arrival times, means, seeds, and policy definitions are saved in a single configuration file.

The first environment has initial means $.65, .50$ and three later batches of inferior arms, so the benchmark remains constant. The second has four successive benchmark levels $.45, .70, .90, .98$; some earlier unresolved membership questions become obsolete when a better arm arrives. The third introduces six arms at one time, including a new best arm. The fourth ends arrivals at pull 4037 and leaves nearly 16,000 further pulls for recovery. Noninitial batches occur 36 pulls after a 100-pull admission grid, so the short-period baseline is not evaluated only at its most favorable alignment. No policy is given future arrivals, the means, or the evaluation horizon.

We implement CP-UCB-AA with FIFO service every second pull at two fixed precisions. The choice $\rho = .04$ satisfies $2\rho < \eta$: on the coverage event, completing all checkpoints resolves every η-separated arm. The choice $\rho = .12$, inherited from the implementation supplied with the theoretical development, is a coarser engineering setting. Its labels remain reliable, but it need not resolve every arm counted by the $\eta = .10$ metric. We report these choices separately. With eight arms their total checkpoint quotas are 73,083 and 8,123 pulls, respectively. The finer setting therefore cannot complete all quotas within the chosen budget.

Three comparisons isolate the allocation rule while retaining the coarser checkpoint report: FIFO service every eighth pull, confidence-driven service every second pull, and full-history UCB without designated service pulls. Confidence-driven service samples the widest interval among unresolved arms and the arm with largest upper endpoint, breaking ties by sample count and then arrival order. It stops designated service when no report remains unresolved. A completed checkpoint retains its fixed interval.

We also use the full-prefix confidence report from the theory with three reward or sampling rules. UCB-prefix admits every arm immediately and uses the same full-history UCB index as CP-UCB-AA. Growing-MOSS uses the anytime index

$$\widehat{\mu}_{i,n} + \sqrt{\frac{\max\{\log(t/(K_t n)), 0\}}{n}},$$

Table 1: Final pseudo-regret and cumulative η-unresolved arm-time at $T = 20{,}000$. Entries are means (Monte Carlo standard errors), from 160 paired replications. Area is in units of 10^3. The two checkpoint precisions have different completeness guarantees.

Scenario	Method	Regret	Area / 10^3
Stable benchmark	CP-FIFO ($\rho = .04$)	1360.5 (1.5)	111.78 (0.00)
	CP-FIFO ($\rho = .12$)	3629.9 (0.7)	49.72 (0.23)
	UCB-prefix	441.7 (2.3)	83.97 (0.46)
	Growing-MOSS	77.3 (0.8)	92.43 (0.00)
	ALL-ε adapt.	1367.8 (11.6)	11.31 (0.09)
	Batch100-UCB	209.3 (2.6)	85.27 (0.52)
Successive improvements	CP-FIFO ($\rho = .04$)	5297.4 (1.2)	74.82 (0.00)
	CP-FIFO ($\rho = .12$)	1194.7 (0.0)	14.74 (0.10)
	UCB-prefix	246.2 (1.7)	18.45 (0.14)
	Growing-MOSS	53.1 (0.8)	33.65 (0.11)
	ALL-ε adapt.	2905.8 (2.4)	9.53 (0.13)
	Batch100-UCB	238.8 (2.1)	22.85 (0.09)
Burst	CP-FIFO ($\rho = .04$)	2798.0 (1.6)	75.89 (0.00)
	CP-FIFO ($\rho = .12$)	3392.2 (0.2)	28.21 (0.17)
	UCB-prefix	354.1 (2.2)	41.75 (0.39)
	Growing-MOSS	69.3 (0.9)	57.03 (0.01)
	ALL-ε adapt.	2017.4 (9.8)	10.31 (0.09)
	Batch100-UCB	278.9 (2.7)	55.87 (0.15)
Arrivals stop	CP-FIFO ($\rho = .04$)	4759.0 (1.3)	89.86 (0.00)
	CP-FIFO ($\rho = .12$)	2645.9 (0.3)	25.14 (0.01)
	UCB-prefix	248.8 (1.5)	45.96 (0.56)
	Growing-MOSS	52.0 (0.6)	71.59 (0.01)
	ALL-ε adapt.	3027.5 (4.3)	12.07 (0.07)
	Batch100-UCB	214.2 (1.8)	55.28 (0.21)

with infinite index at $n = 0$; no minimax property of the fixed-horizon MOSS algorithm is assumed for this extension. ALL-ε adaptation uses the confidence-driven service rule every second pull and UCB otherwise. It retains all history and can revise positive reports after arrivals. This is a natural adaptation of confidence-based all-good-arm sampling, not an implementation of FAREAST or $(\mathrm{ST})^2$. Finally, Batch100-UCB follows the reviewer's proposed short-admission comparison: it admits waiting arms at pulls $1, 101, 201, \ldots$, uses UCB on the admitted reward-active arms, and retires an arm from reward sampling when its upper bound falls below the largest active lower bound. It retains all observations. Its report covers all physically arrived arms, including those still waiting for admission, whose intervals are $[0, 1]$.

Table **??** gives means and Monte Carlo standard errors. Figure **??** includes every method; error bars are 1.96 Monte Carlo standard errors. The released data also contain paired differences, all unresolved paths, service pulls, and the regret charged to service. The compiled implementation was checked against the original CP-UCB-AA `choose/observe/report` class for 20,000 successive rounds: actions, service indicators, every available-arm count, and every reported label agreed at each round.

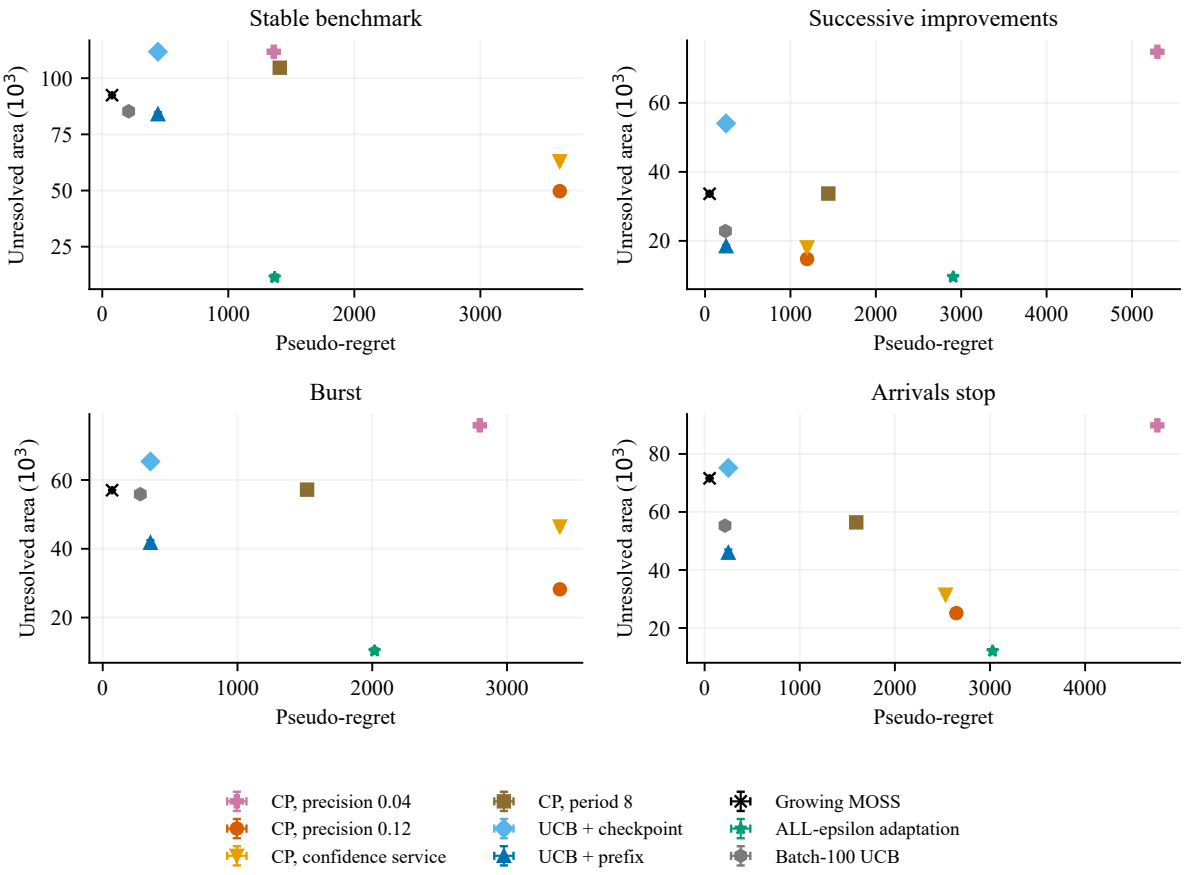


Figure 1: Regret and cumulative η-unresolved area for all nine methods. Lower values on both axes are preferable. Error bars quantify Monte Carlo uncertainty, not uncertainty in the theoretical reliability statement. Neither checkpoint precision nor its allocation variants uniformly dominate the confidence-based adaptation.

6.2 What the comparisons show

The finer checkpoint rule leaves between five and eight η-unresolved arms on average at the final pull, depending on the environment. Its unresolved area ranges from 74,820 to 111,784. This is an observable limitation of the conservative construction, rather than evidence that arrivals are intrinsically unmanageable at this budget. The confidence-based adaptation resolves all η-separated arms by the end in every replication of these four environments.

The coarser checkpoints reduce area substantially relative to UCB alone, but can incur much larger regret. Under a stable benchmark, CP-FIFO has mean regret 3629.9 and area 49,718, compared with 1367.8 and 11,306 for ALL-ε adaptation. The paired differences, CP-FIFO minus the adaptation, are 2262.1 (standard error 11.7) for regret and 38,411.7 (standard error 241.9) for area. The adaptation also improves both quantities in the burst environment. These comparisons do not support a general numerical advantage of checkpoint FIFO over a strong history-retaining identification rule.

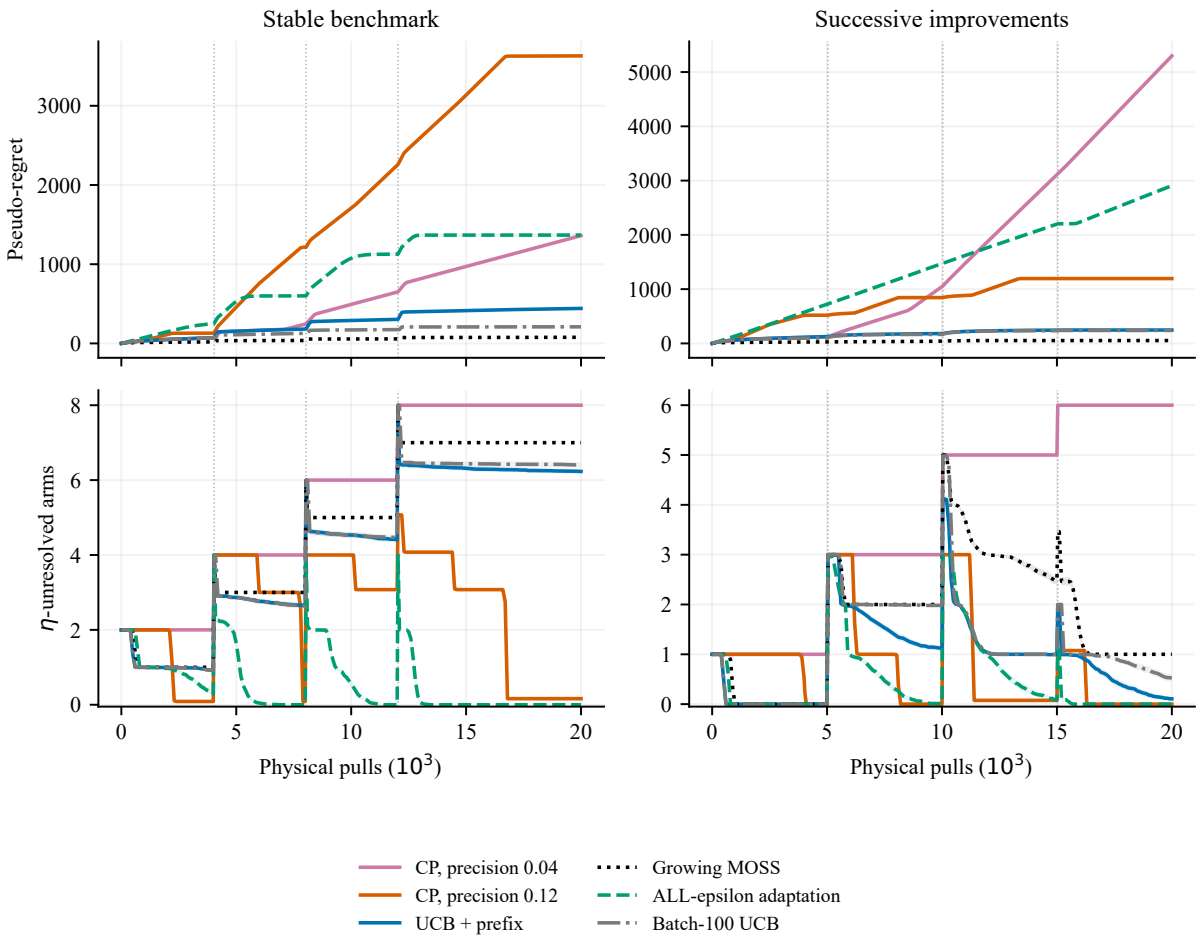


Figure 2: Mean regret and mean η-unresolved arm counts at the physical sampling clock. Bands are 1.96 Monte Carlo standard errors; vertical dotted lines mark arrivals. Stable and improving benchmarks are shown here; the other environments appear in the supplement. The main six methods are displayed; all nine methods and all sampled paths are retained in the released results.

In the successive-improvement environment, CP-FIFO instead occupies a different cost point: regret 1194.7 and area 14,743, compared with 2905.8 and 9525 for the confidence-based adaptation. Here the paired regret difference is -1711.0 (standard error 2.4), while the area difference is 5217.7 (standard error 141.1). This is consistent with the distinction between spending pulls to resolve current membership questions and allowing some questions to be superseded by later arrivals. It is not an estimate of an optimal Pareto frontier.

Growing-MOSS has the smallest mean regret in all four environments, between 52.0 and 77.3, while leaving substantially larger unresolved areas. Low regret therefore provides little evidence of timely set maintenance. Conversely, the explicit service rules pay for observations of inferior arms. Reducing service from every second to every eighth pull generally reduces regret and enlarges unresolved area. Replacing FIFO by the confidence-driven checkpoint allocation does not uniformly improve either criterion; allocation and the report's precision both matter.

For Batch100-UCB we additionally record the exact decomposition into within-active regret, admission delay, and loss from previous reward retirement. Its admission-delay contributions are 0, 33.92, 16.00, and 22.40 in the four environments. No historical-retirement deficit occurred in these runs. The largest discrepancy between total regret and the sum of the three recorded components was below 3.1×10^{-12}. The corresponding admission and retirement terms are identically zero for the methods that admit every arm immediately and retain every arm for reward selection. This accounting avoids attributing every loss outside within-active learning to arrival delay.

There were no incorrect positive or negative reports in any of the 5760 runs. For one fixed environment and method, zero failures in 160 replications gives a one-sided 95% upper confidence bound of 0.0186 on its failure probability. This finite set of diagnostics does not establish the uniform, all-environment reliability guarantee. Nor do these small eight-arm experiments validate the asymptotic lower bounds, resolve the critical logarithmic gap, or establish checkpoint optimality. They show the practical cost of the conservative quotas and the importance of comparing retained-history methods under both regret and maintenance criteria.

7 Discussion

Reliable maintenance of a relative near-optimal set imposes a cost that is not visible from allocation regret alone. One arrival can affect many reports through uncertainty about the common reference level. Repeated improvements can also make fine discrimination of earlier candidates unnecessary. The cumulative area criterion measures both effects in the same experiment and states how much unresolved time a reporting requirement permits.

The strength of the timeliness requirement matters. Sublinear area here means that the average unresolved *count* vanishes. An application concerned only with the unresolved proportion, the discovery of one good arm, or a terminal decision would face a different task. The arrival exponent and quadratic worst-case area depend on this normalization. Arms in the boundary band remain in the model: their unresolved time is exempt from the loss, while every issued label remains subject to correctness.

The finite-time results and eventual consistency also answer different questions. If arrivals stop, the final arm set is finite, and every arm is strictly separated from the membership boundary, a revisable multiresolution version eventually reports the exact set almost surely and has sublinear expected regret. Its construction and proof are supplied in the supplement. For the fixed-precision checkpoint implementation, frozen intervals can remain too wide and an early reporting error need not be corrected, so this consistency conclusion does not apply. The reward layer retains every available arm and can still correct allocation.

Our bounds leave a substantive instancewise question. Under arbitrary overlapping arrivals, heterogeneous boundary distances, and intermediate area budgets, current information constraints need not describe an attainable causal allocation. A matching theory would have to price common reference evidence, unfinished tasks, their possible expiration, and observations already obtained for reward collection without charging the same observation twice. The restricted multiple-update

theorem leaves that scheduling problem open. The critical logarithmic gap on an individual smooth schedule and automatic choice of service precision also remain open.

For practice, the numerical evidence favors examining the two costs together and comparing methods that retain their history. CP-UCB-AA gives transparent guarantees, but fixed checkpoints can devote many observations to tasks already unimportant at a moderate horizon. Confidence-driven adaptations can be much more efficient, motivating sharper continual-service guarantees for those rules. A smaller reporting set alone does not establish a resource advantage.

Supplementary material and reproducibility. The supplement contains complete proofs, the information-allocation lower relaxation, the quantile extension, the revisable consistency construction, and further allocation diagnostics. The numerical package contains the policy definitions, frozen configurations, every replication, paired Monte Carlo summaries, figure code, and a source-class parity check. All numerical data in this article are simulated; no patient-level or other original human-subject data were used.

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